

Manoj Chapagain

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SUMMARY STATEMENT

Senior Software Engineer with 8+ years of experience designing and scaling distributed systems that power high-impact products across fintech, cybersecurity, IoT, and SaaS. Recognized for driving complex projects from concept to production and consistently delivering solutions that enhance reliability, performance, and user experience at scale.

EXPERIENCE

Mechatronic

Canada

Software Engineer

Feb. 2024 – Present

- Owned full lifecycle of projects from code to production-grade, leveraging Terraform, Kubernetes, and automated CI/CD pipelines
- Built firmware module for resource-constrained IoT devices and responsive React web interface, enabling real-time device management at scale
- Designed, fine-tuned, and deployed AI agents and automation workflows with LangGraph, OpenAI API, and Pinecone, slashing manual ops

Microsec

Singapore

Backend Developer

May. 2022 – Apr. 2023

- Engineered PKI security protocols from concept to deployment on IoT/OT devices, using Python Cryptography library for certificate provisioning, encrypted messaging, and integrity
- Architected microservices from design to high-availability clusters, deploying Docker + Kubernetes across multi-node setups with high-availability and fast auto-scaling

Codavatar

Kathmandu, Nepal

Sr. Software Engineer

Jan. 2021 – Mar. 2022

- Led backend team from roadmap to on-time delivery of production services, hitting every milestone under tight deadlines
- Optimized Krispcall cloud telephony from core features to 50 k concurrent calls, integrating WebRTC, Twilio, FastAPI, and Redis
- Refactored codebase with clean architecture; applying domain-driven design to cut technical debt and improve API performance

Full Stack Developer

Nov. 2017 – Dec. 2020

Janaki Tech.

Lalitpur, Nepal

- Built fintech frontend from zero to 100 k users in 12 months, scaling React app for early-stage startup
- Engineered PCI-compliant payment gateway and digital wallet from backend to live integrations, using Python/Django, React, and AWS (Lambda, RDS, S3) with 50+ bank/telecom APIs

Jr. Full-Stack Developer

Dec. 2016 – Nov. 2017

GrepSr

Kathmandu, Nepal

- Developed web-scraping pipelines and internal projects; using microservices, gRPC, Docker, and Kubernetes

EDUCATION

Seneca College

Toronto, Canada

Project Managemnt - IT

May. 2023 – Dec. 2023

Tribhuvan University, IOE Pulchowk Campus

Lalitpur, Nepal

Bachelor's degree in Computer Engineering

Nov. 2012 – Sep. 2016

PUBLICATIONS

Effects of Auto-Tuning and Pitch Normalization on Query by Humming

June 2020

International Journal of Advanced engineering

Source

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Shell (bash/zsh)

Frameworks & Libraries: FastAPI, Django/DRF, Flask, React, Node.js, gRPC, GraphQL (Ariadne/Graphene), SQLAlchemy, Redis, RabbitMQ, LangGraph, OpenAI API, Pinecone, scikit-learn, Pandas, Selenium

DevOps & Cloud: Kubernetes (K8s), Docker, Terraform, AWS (EC2, S3, RDS, Lambda), CI/CD (GitHub Actions, GitLab CI), Ansible, Nginx, Prometheus, Grafana

Testing & Docs: Pytest, Cypress, Postman, Swagger/OpenAPI, Ruff, Flake8, MkDocs

Others: Linux, Jira, Asana